JOYSTICK

+

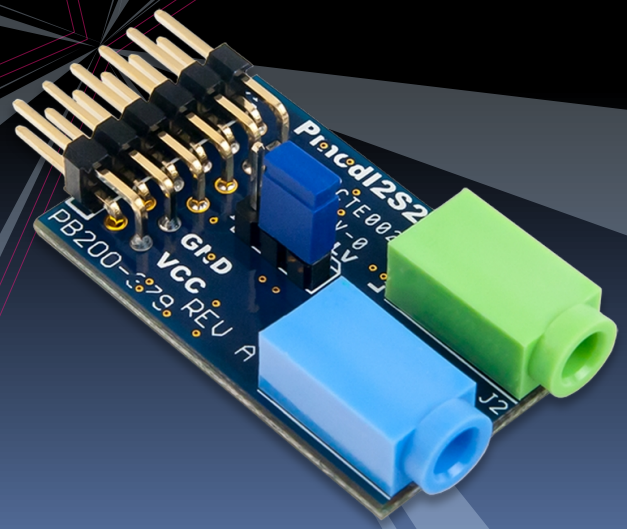
SPI

Hardware modules

At the beginning of LAB2, we will give each one of you a Digilent Pmod Joystick SPI module and Digilent Pmod I2S2 module.

This can be connected to your Basys3 board through the Pmod connectors.



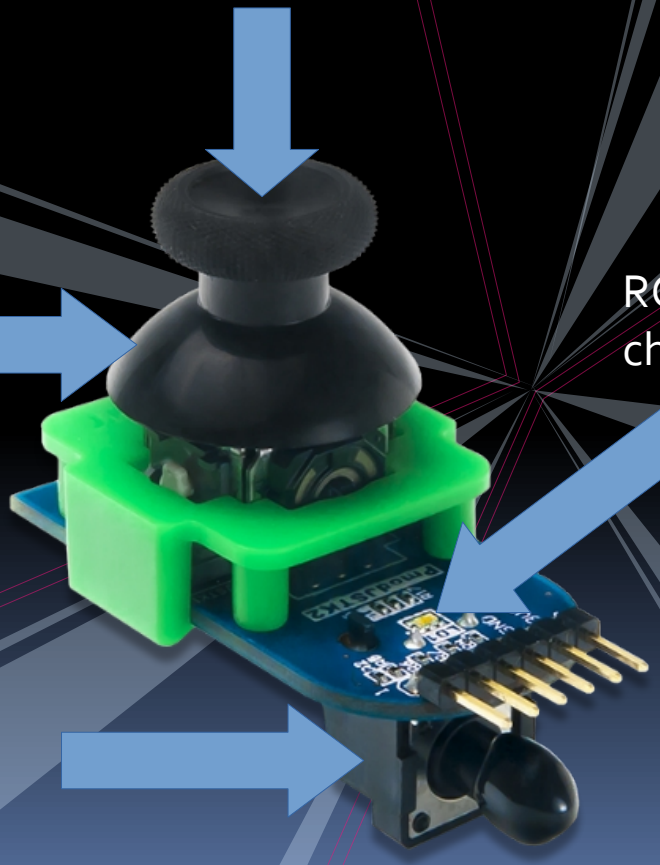


Digilent Pmod JSTK2 components

Joystick button (1 bit, pressed/not-pressed)

Joystick position (10 bits X-axis, 10 bits Y-axis)

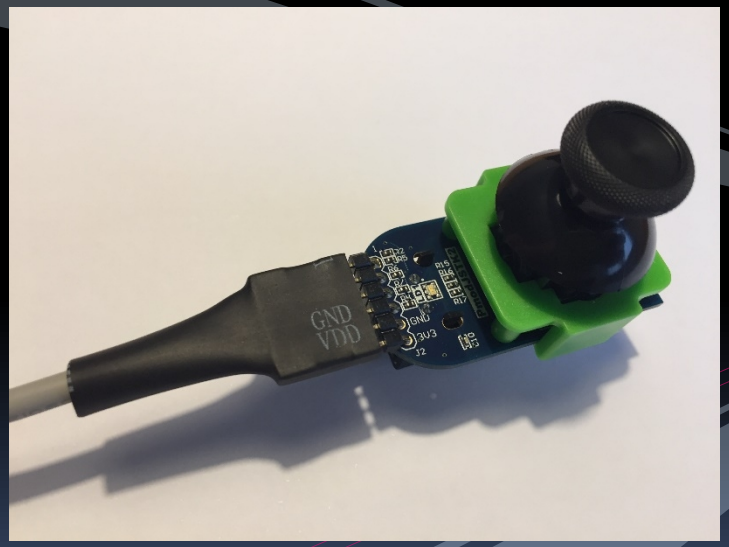
RGB LED (8 bits for each channel)

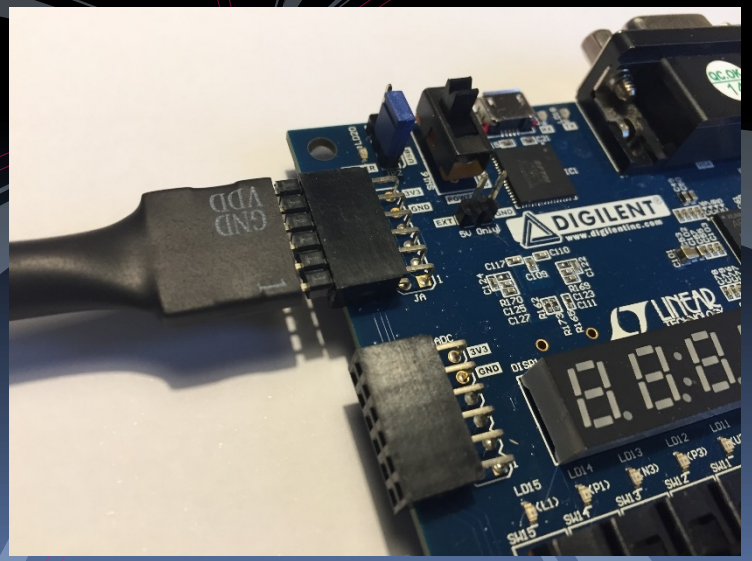


"Trigger" button (1 bit, pressed/not-pressed)

How to connect the Joystick

Connect the Joystick with the provided cable to "JA", top row, paying attention to the VCC and GND position.





JSTK2 interfacing

The Digilent Pmod JSTK2 module protocol is fully described in its reference manual.

In short, it uses the SPI protocol to receive "commands" and send back the "readings" of the Joystick position and the buttons state.

SPI protocol

The Serial Peripheral Interface (SPI) is a very popular synchronous protocol for off-chip communication.

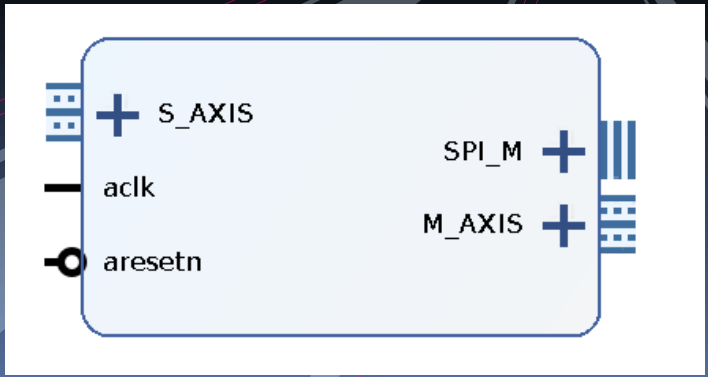
In its basic form it is composed by 4 signals:

- Serial CLocK (SCLK)
- Master-Out Slave-In (MOSI)
- Master-In Slave-Out (MISO)
- Chip Select (CS)

SPI IP-Core

While SPI is a simple protocol, describing it in VHDL is not immediate.

To ease your work, we will give you a "AXI4-Stream SPI" IP-Core, similar to the UART one.



SPI IP-Core

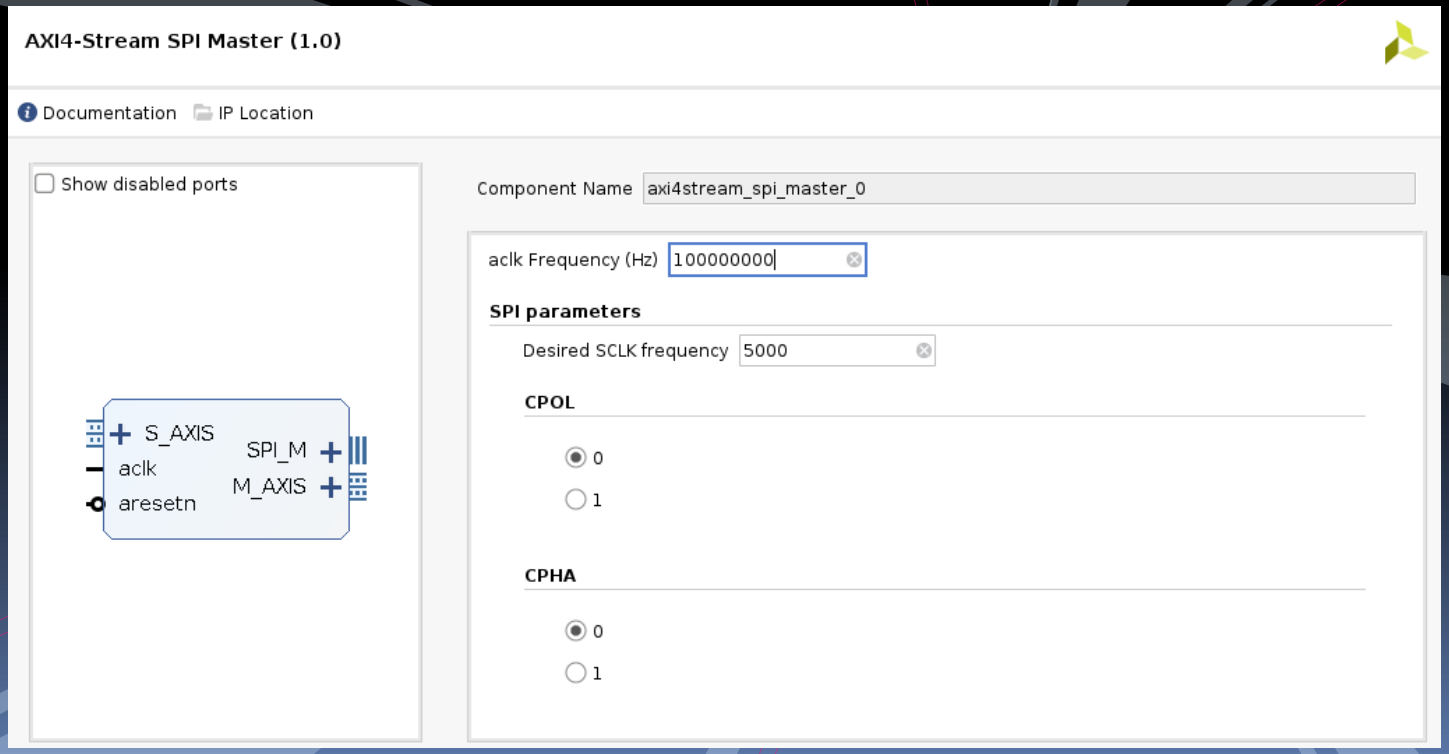
You will find the details of its behavior in the README file. In short:

- Whatever you send to S_AXIS is sent to the SPI "slave".
- Whatever is received from the SPI "slave" is sent to you through the M_AXIS interface.



SPI IP-Core

To respect the timing of the Digilent Pmod JSTK2 module (see here), use these parameters in the SPI IP-Core.



JSTK2 commands

Using this IP-Core, you can easily "talk" to the module, following the protocol described in the reference manual.

We suggest you to use the cmdSetLedRGB command so that, in a single command, you can get the position of the joystick and the state of the button, and control the LED color.

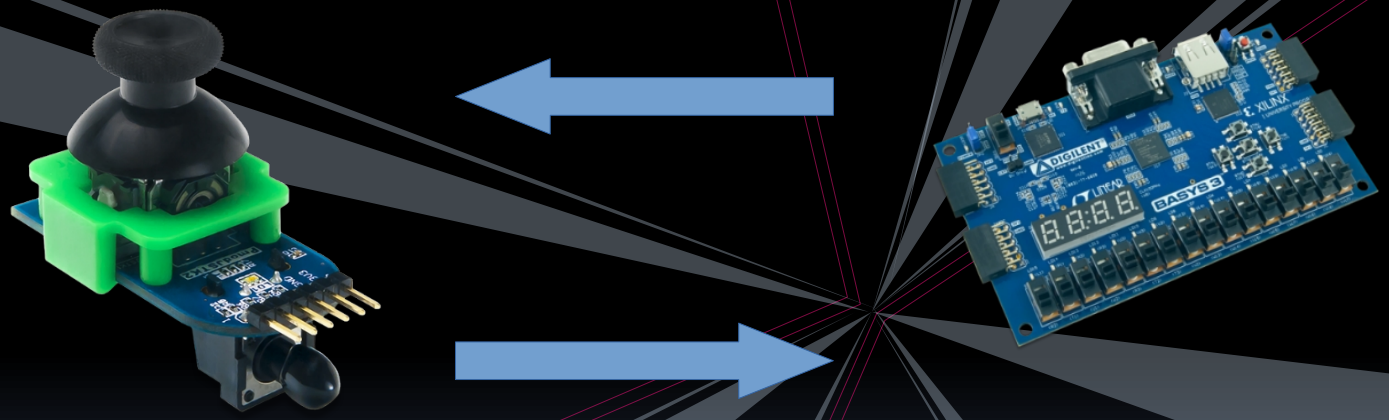
JSTK2 commands - cmdSetLedRGB

		Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
cmdSetLedRGB	**(0x84)**	1	0	0	0	0	1	0	0

Parameters

PARAM1 – Red LED duty cycle
PARAM2 – Green LED duty cycle
PARAM3 – Blue LED duty cycle
PARAM4 – ignored

Set the duty cycles for the Red, Green, and Blue LEDs.



	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5
MOSI	COMMAND / 0	PARAM1 / DUMMY	PARAM2 / DUMMY	PARAM3 / DUMMY	PARAM4 / DUMMY
MISO	smpX (Low Byte)	smpX (High Byte)	smpY (Low Byte)	smpY (High Byte)	fsButtons

	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
fsButtons	EXTPKT	0	0	0	0	0	TRIGGER	JOYSTICK

EXTPKT: Extended Packet Status Bit
1 = additional data corresponding to the command byte is available and may be retrieved after this byte
0 = standard response packet, no additional data follows this byte

TRIGGER: Trigger Button Status Bit
1 = trigger button is currently pressed
0 = trigger button is not being pressed

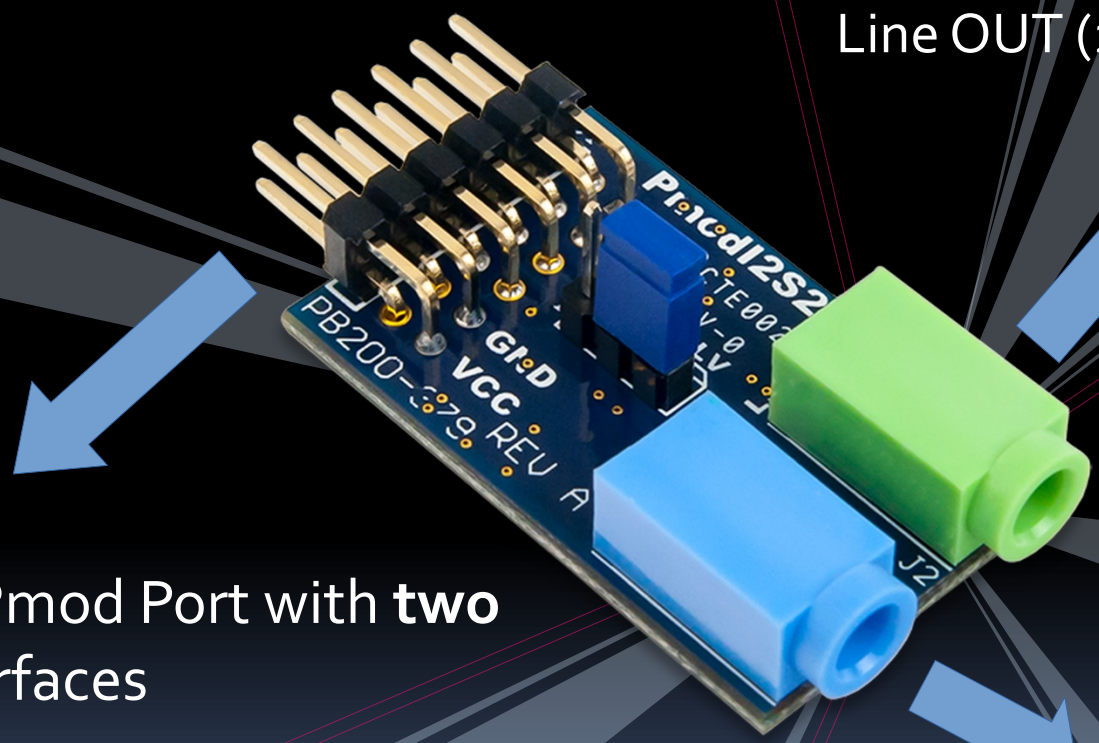
JOYSTICK: Joystick Center Button Status Bit
1 = joystick center button is currently pressed
0 = joystick center button is not being pressed

AUDIO

+

I2S

Digilent Pmod I2S2 components



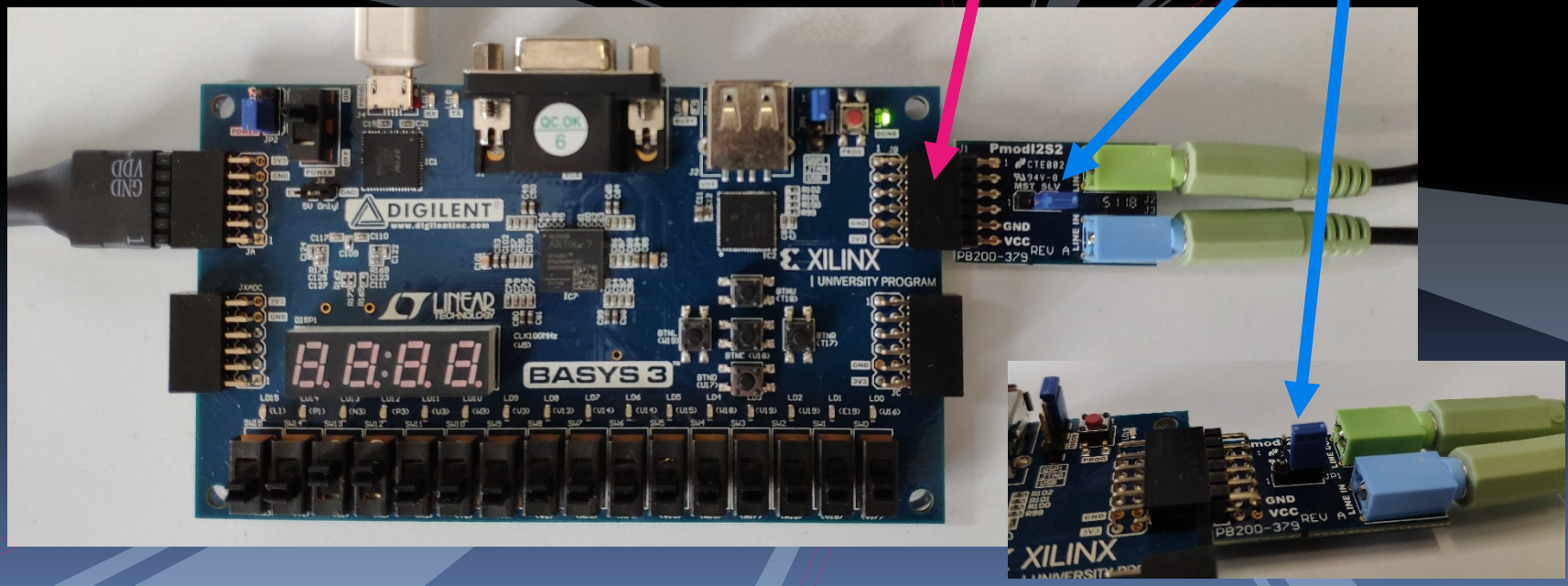
Line OUT (24 bit D/A)

12-pin Pmod Port with **two** I2S interfaces

Line IN (24 bit A/D)

How to connect the I2S2 module

The provided constraints are for the JB connector (top right of the board). Also make sure that the jumper on the Pmodi2S2 module is on the **SLV** position (right position).



Pmod I2S2 interfacing

The Digilent Pmod I2S2 module protocol is fully described in its reference manual.

In short, it uses the I2S protocol for both receiving an input audio signal from a source (through an ADC) and sending back an audio signal by means of a DAC to any kind of "speaker".

Audio signal format

- Typical sound frequency passband of the ears is between 20 and 20KHz.
  (Shannon theorem => 44.1 Ksamples)

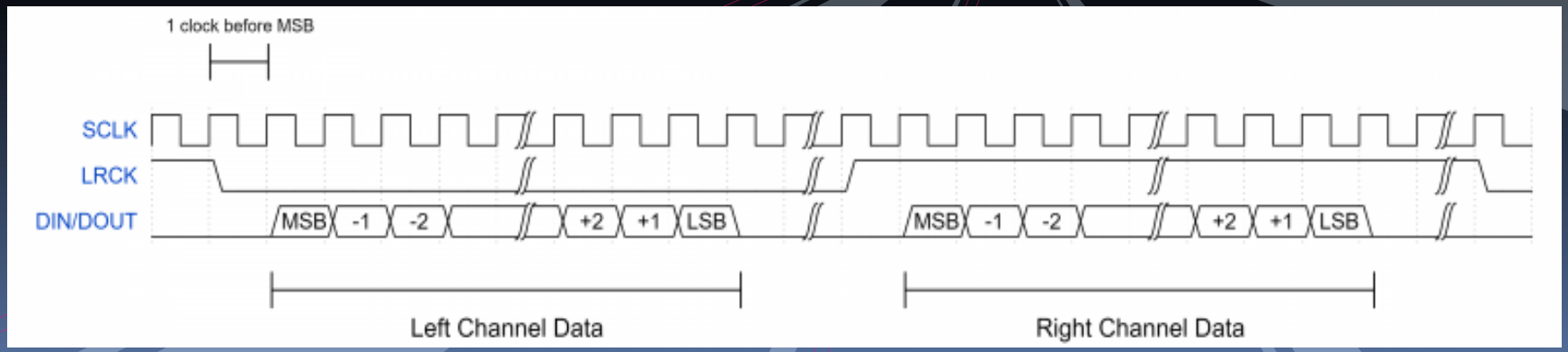
- For a good quality audio we choose a 24bits/channel depth.
  (we consider stereo audio: L/R channels)

I2S protocol

The Inter-IC Sound (I2S) is a popular serial protocol for digital audio devices connections.

In its basic form it is composed by 4 signals:

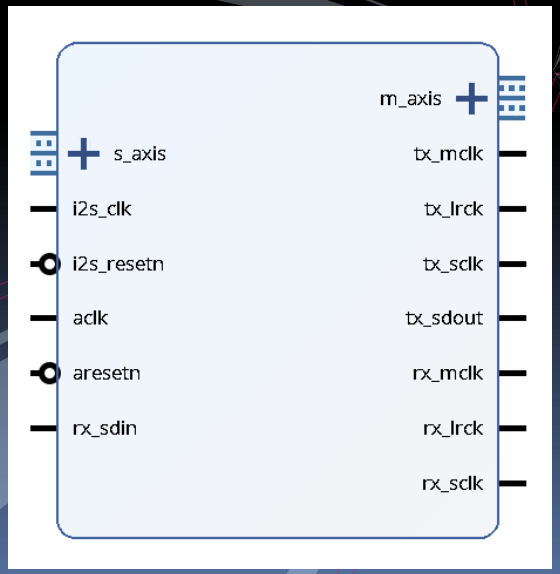
- Master CLocK (MCLK)
- Left/Right CLocK (LRCLK, aka Word Select)
- Serial CLocK (SCLK, aka Bit Clock)
- Din/Dout (for Line In/Line Out channels)



I2S IP-Core

While I2S is a simple protocol, describing it in VHDL is not immediate.

To ease your work, we will give you a "AXI4-Stream I2S2" IP-Core, similar to the SPI and UART ones.



I2S IP-Core

The provided Pmod-I2S2 IP-Core has:

- Two I2S interfaces, to be connected to the external pins of the FPGA.
- Two AXI4-Stream interfaces, to read the data from the ADC or to send the data to the DAC.
- Two clocks:
  - one for the I2S and (22.579 MHz)
  - one for the AXI4-Stream
- Two input active-low reset signals (one for each clock domain).

I2S IP-Core: 44.1 kHz Audio

The IP core needs a 22.579 MHz clock (to be connected to i2s_clk) and a 100 MHz clock (to be connected to every other clock input).

In details, for a 44.1 kHz audio sampling rate:

MCLK = 22.579 MHz (I2S Clk of the IP-Core)

SCLK = MCLK/8

Pmod-I2S2 AXI4-Stream format

The AXI4-Stream interface has an additional line called **TLAST**, which is used to determine the end of a packet.

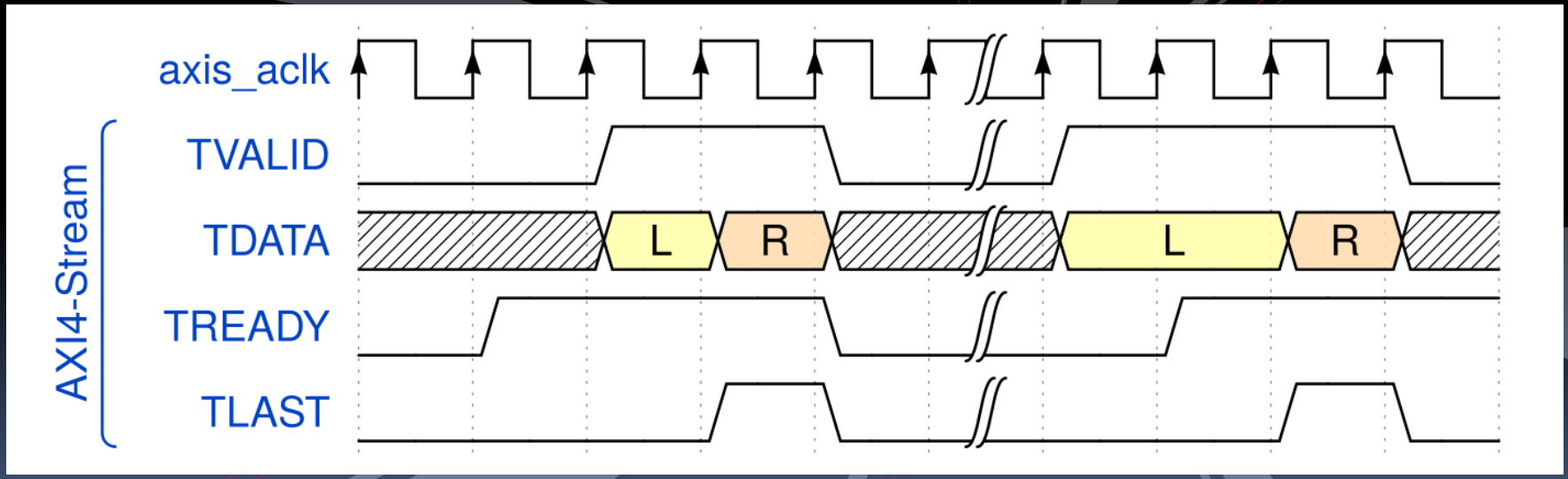
Each packet is composed by two 24-bits words: the first one is the audio data of the left channel, the second one the audio data of the right one.

TLAST is asserted on the second word; in other words:

- TLAST = 0: left channel
- TLAST = 1: right channel

Pmod-I2S2 AXI4-Stream format

In this example, two «packets» have been transferred, first the left channel and then the right one for each one.



First step: loopback test

